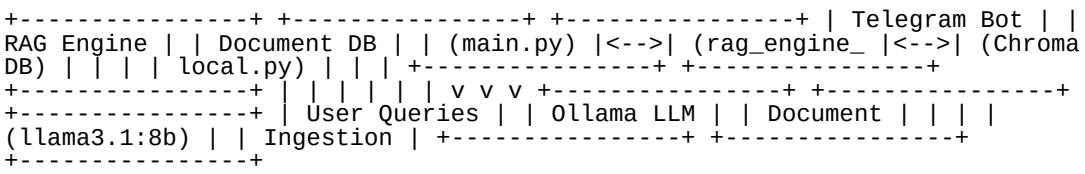
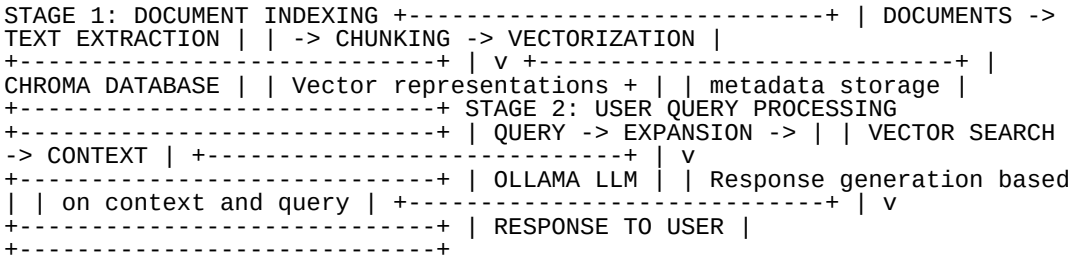
Kcell RAG Bot - System Diagrams

System Architecture Diagram



System Workflow Diagram



System Components

SYSTEM COMPONENTS: 1. TELEGRAM BOT - main.py - main bot file - config.py - configuration - Command and message processing 2. RAG ENGINE - rag_engine_local.py - RAG engine - Ollama integration - Query processing 3. VECTOR DATABASE - Chroma DB - SentenceTransformers embeddings - Document metadata 4. DOCUMENT INGESTION - ingest.py - document indexing - Format support: PDF, DOCX, XLSX, PPTX - OCR for scanned documents 5. AI MODELS - Ollama (local LLM) - llama3.1:8b - main model - gemma:2b - backup model